SECTION 04 05 03

MASONRY MORTARING AND GROUTING

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Mortar and grout for masonry.

1.02 RELATED SECTIONS

A. Section 01 40 00 - Quality Requirements.

B. Section 04 20 16 - Reinforced Unit Masonry.

1.03 REFERENCES

A. ASTM C91 - Masonry Cement.

B. ASTM C94 - Ready-Mixed Concrete.

C. ASTM C144 - Aggregate for Masonry Mortar.

D. ASTM C150 - Portland Cement.

E. ASTM C207 - Hydrated Lime for Masonry Purposes.

F. ASTM C270 - Mortar for Unit Masonry.

G. ASTM C404 - Aggregates for Masonry Grout.

H. ASTM C476 - Grout for Masonry.

I. ASTM C780 - Preconstruction and Construction Evaluation of Mortars for Plain and Reinforced Unit Masonry.

J. ASTM C1019 - Method of Sampling and Testing Grout.

K. IMIAC (International Masonry Industry All-Weather Council) -Recommended Practices and Guide Specifications for Cold Weather Masonry Construction.

L. ACI 530 – Building Code Requirements for Masonry Structures.

M. ACI 530.1 – Specification for Masonry Structures.

1.04 SUBMITTALS

A. Submit under provisions of Section 01 33 00.

B. Include design mix, indicate whether the Proportion or Property specification of ASTM C270 is to be used, required environmental conditions, and admixture limitations.

C. Reports: Submit reports on mortar indicating conformance of mortar to property requirements of ASTM C270 and test and evaluation reports to ASTM C780.

D. Reports: Submit reports on grout indicating conformance of component grout materials to requirements of ASTM C476 and test and evaluation reports to ASTM C1019.

E. Manufacturer's Certificate: Certify that products meet or exceed specified requirements.

1.05 QUALITY ASSURANCE

A. Maintain one copy of each document on site.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, protect, and handle products to site under provisions of Section 01 60 00.

B. Maintain packaged materials clean, dry, and protected against dampness, freezing, and foreign matter.

1.07 ENVIRONMENTAL REQUIREMENTS

A. Cold Weather Requirements: IMIAC - Recommended Practices and Guide Specifications for Cold Weather Masonry Construction.

PART 2 PRODUCTS

2.01 MATERIALS

A. Portland Cement: ASTM C150, Type V

B. Masonry Cement: ASTM C91, Type M OR S. Type V cement.

C. Mortar Aggregate: ASTM C144, standard masonry type.

D. Grout Course Aggregate: ASTM C404.

E. Water: Clean and potable.

2.02 MORTAR COLOR

A. As indicated on Architectural plans and specifications.

2.03 MORTAR MIXES

A. Mortar For Load Bearing Walls: ASTM C270, Type M or S using the Property Method.

B. Pointing Mortar: ASTM C270, Type O using the Property Method, with maximum 2 percent ammonium stearate or calcium stearate per cement weight.

2.04 MORTAR MIXING

A. Thoroughly mix mortar ingredients in accordance with ASTM C270 in quantities needed for immediate use.

B. Maintain sand uniformly damp immediately before the mixing process.

C. Do not use anti-freeze compounds to lower the freezing point of mortar.

D. If water is lost by evaporation, re-temper only within two hours of mixing.

E. Use mortar within two hours after mixing at temperatures of 90 degrees F (32 degrees C), or two-and-one-half hours at temperatures under 50 degrees F.

2.05 GROUT MIXES

A. Bond Beams: 2,000 psi compressive strength at 28 days; 8-10 inches slump; mixed in accordance with ASTM C476 Course grout.

B. Engineered Masonry: 2,000 psi strength at 28 days; 8-10 inches slump; mixed in accordance with ASTM C476 Course grout.

2.06 GROUT MIXING

A. Thoroughly mix grout ingredients in quantities needed for immediate use in accordance with ASTM C476 Course grout.

B. Add admixtures in accordance with manufacturer's instructions; mix uniformly.

C. Do not use anti-freeze compounds to lower the freezing point of grout.

PART 3 EXECUTION

3.01 EXAMINATION

A. Request inspection of spaces to be grouted.

3.02 PREPARATION

A. Plug clean-out holes with block masonry units. Brace masonry for wet grout pressure.

3.03 INSTALLATION

A. Install mortar and grout to requirements of the specific masonry sections.

B. Work grout into masonry cores and cavities to eliminate voids.

C. Do not displace reinforcement while placing grout.

D. Remove excess mortar from grout spaces.

END OF SECTION

SECTION 04 20 16
REINFORCED UNIT MASONRY

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Concrete masonry units.

B. Reinforcement, anchorage, and accessories.

1.02 RELATED SECTIONS

A. Section 01 40 00 - Quality Requirements

B. Section 04 05 03 – Masonry Mortaring and Grouting

C. Section 07 62 00 - Sheet Metal Flashing and Trim: Cap flashings over masonry work and placement of reglets for flashings.

D. Section 07 90 00 - Joint Protection: Rod and sealant at control and expansion joints.

E. Section 07 19 00 - Water Repellents

1.03 REFERENCES

A. ASTM A123 - Zinc (Hot-Dipped Galvanized) Coatings on iron and Steel Products.

B. ASTM A615 - Deformed and Plain Billet Steel Bars for Concrete Reinforcement.

C. ASTM C90 - Load-Bearing Concrete Masonry Units.

D. IMIAC - International Masonry Industry All-Weather Council: Recommended Practices and Guide Specification for Cold Weather Masonry Construction.

1.04 SUBMITTALS

A. Submit under provisions of Section 01 33 00.

B. Shop Drawings: Indicate bars sizes, spacing, locations, reinforcement quantities, bending and cutting schedules, supporting and spacing devices for reinforcement and accessories.

C. Product Data: Provide data for masonry units and reinforcement.

D. Samples: Submit four samples of all block units to illustrate color, texture and extremes of color range.

E. Indicate required mortar strength, masonry unit assembly strength in all planes, and supportive test data.

F. Manufacturer's Certificate: Certify that Products meet or exceed specified requirements.

1.05 QUALITY ASSURANCE

A. Maintain one copy of each document on site.

1.06 QUALIFICATIONS:

A. Manufacturer: Company specializing in manufacturing the Products specified in this section with minimum five years documented experience.

1.07 REGULATORY REQUIREMENTS

A. Conform to the 2024 Edition of the International Building Code and all other applicable codes for requirements for masonry construction.

1.08 MOCKUP

A. Provide mockup of masonry under provisions of Section 01 40 00.

B. Construct a masonry wall 8' x 8' which includes mortar and accessories.

C. Locate where directed.

D. Mockup may not remain as part of the Work.

1.09 DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, protect and handle products to site under provisions of Section 01 60 00.

1.10 ENVIRONMENTAL REQUIREMENTS

A. Cold Weather Requirements: IMIAC - Recommended Practices and Guide Specifications for Cold Weather Masonry Construction.

B. Maintain materials and surrounding air temperature to maximum 90 degrees F (32 degrees C) prior to, during, and 48 hours after completion of masonry work.

1.11 COORDINATION

A. Coordinate work under provisions of Section 01 30 00.

1.12 EXTRA MATERIALS

A. Submit under provisions of Section 01 33 00.

B. Provide 50 of each size, color, and type of units.

PART 2 PRODUCTS

2.01 MANUFACTURERS

A. Manufacturer of concrete masonry units to be approved by Owner.

2.02 CONCRETE MASONRY UNITS

A. Hollow Load Bearing Block Units (CMU): ASTM C90, Grade N, Type II/V cement, Type I -Moisture Controlled; Medium weight; 2000 psi minimum compressive strength.

B. Provide special units for 90 degree corners, bond beams and lintels.

2.03 REINFORCEMENT AND ANCHORAGE

A. Reinforcing Steel: ASTM A615, 60 ksi yield grade, deformed billet bars as indicated on the project plans.

2.04 MORTAR AND GROUT

A. As specified in Section 04 05 03.

2.05 FLASHINGS

A. Galvanized Steel: ASTM A525, G90 finish, 20 gage core steel.

2.06 ACCESSORIES

A. Preformed Control Joints: Neoprene material. Provide with corner and tee accessories.

B. Joint Filler: Closed cell, oversized 50 percent to joint width; self expanding.

C. Building Paper: No. 15 asphalt saturated felt.

D. Cleaning Solution: Non-acidic, not harmful to masonry work or adjacent materials.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that field conditions are acceptable and are ready to receive work.

B. Verify items provided by other sections of work are properly sized and located.

Verify that built-in items are in proper location, and ready for roughing into masonry work.

3.02 PREPARATION

A. Direct and coordinate placement of metal anchors supplied to other Sections.

B. Provide temporary bracing during installation of masonry work. Maintain in place until building structure provides permanent bracing.

3.03 COURSING

A. Establish lines, levels, and coursing indicated. Protect from displacement.

B. Maintain masonry courses to uniform dimension. Form vertical and horizontal joints of uniform thickness.

C. Bond: Running.

D. Coursing: One unit and one mortar joint to equal 8 inches unless indicated otherwise on drawings.

E. Mortar Joints: Match existing CMU at site.

3.04 PLACING AND BONDING

A. Lay solid masonry units in full bed of mortar, with full head joints, uniformly jointed with other work.

B. Lay hollow masonry units with face shell bedding on head and bed joints.

C. Buttering corners of joints or excessive furrowing of mortar joints are not permitted.

D. Remove excess mortar as work progresses.

E. Interlock intersections and external corners.

F. Do not shift or tap masonry units after mortar has achieved initial set. Where adjustment must be made, remove mortar and replace.

G. Perform job site cutting of masonry units with proper tools to provide straight, clean, unchipped edges. Prevent broken masonry unit corners or edges.

H. Cut mortar joints flush where wall tile is scheduled.

I. Isolate masonry partitions from vertical structural framing members with a control joint as indicated.

J. Isolate top joint of masonry partitions from horizontal structural framing members and slabs or decks.

K. Do not place chipped or broken concrete masonry units.

3.05 REINFORCEMENT AND ANCHORAGE

A. Install horizontal reinforcement as indicated.

B. Place masonry reinforcement in first horizontal course above and below openings. Extend minimum 24 inches each side of opening.

C. Place reinforcement continuous in first course below top of walls.

D. Lap reinforcement ends as indicated on the project plans.

E. Support and secure reinforcing bars from displacement. Maintain position within 1/2 inch of dimensioned position.

F. Embed anchors as indicated.

3.06 MASONRY FLASHINGS

A. Extend flashings under veneer, turn up minimum 8 inches and bed into mortar joint of masonry seal to concrete over framed back-up.

B. Lap end joints minimum 6 inches and seal watertight.

C. Use flashing manufacturer's recommended adhesive and sealer.

3.07 LINTELS

A. Use single piece reinforcing bars only.

B. Support and secure reinforcing bars from displacement. Maintain position within 1/2 inch of dimensioned position.

C. Place and consolidate grout fill without displacing reinforcing.

D. Allow masonry lintels to attain specified strength before removing temporary supports.

E. Maintain minimum 8 inch bearing on each side of opening.

3.08 GROUTED COMPONENTS

A. Lap Splices: Lap reinforcing bars in accordance with the requirements of the 2018 Edition of the International Building Code and the project plans.

B. Support and secure reinforcing bars from displacement. Maintain position within 1/2 inch of dimensioned position.

C. Place and consolidate grout fill without displacing reinforcing.

D. At bearing locations, fill masonry cores with grout for a minimum of 16 inches either side of opening.

3.09 ENGINEERED MASONRY

A. Lay masonry units with core cells vertically aligned and unobstructed.

B. Place mortar in masonry unit bed joints back 1/4 inch from edge of unit grout spaces. Permit mortar to cure 2 days before placing grout.

C. Reinforce masonry unit cores with reinforcement bars and grout as indicated.

D. Retain vertical reinforcement in position at top and bottom of cells and at intervals not exceeding 192 bar diameters. Splice reinforcement in accordance with this Section.

E. Grout spaces less than 2 inches in width with fine grout using low lift grouting techniques. Grout spaces 2 inches or greater in width with course grout using high or low lift grouting techniques.

F. When grouting is stopped for more than one hour, terminate grout 1-1/2 inches below top of upper masonry unit to form a positive key for subsequent grout placement.

3.10 CONTROL AND EXPANSION JOINTS

A. Do not continue horizontal joint reinforcement through control and expansion joints except at roofs and top of wall.

B. Install preformed control joint device in continuous lengths. Seal butt and corner joints in accordance with manufacturer's instructions.

C. Size control joint in accordance with Section 07 90 00 for sealant performance.

D. Form expansion joint as detailed.

E. Place control joints in locations approved by Engineer and shown on plans.

F. Unless noted otherwise on the project plans, place control joints such that maximum spacing between joints does not exceed 40 feet nor twice the height of the wall.

G. Do not place control joints within 4 feet of any opening in wall.

3.11 BUILT-IN WORK

A. As work progresses, install built-in metal door frames, fabricated metal frames, window frames, fireplace accessories, anchor bolts, plates, and other items to be built-in the work and furnished by other sections.

B. Install built-in items plumb and level.

C. Bed anchors of metal door frames in adjacent mortar joints. Fill frame voids solid with grout.

D. Do not build in organic materials subject to deterioration.

3.12 TOLERANCES

A. Maximum Variation From Alignment of Pilasters: 1/4 inch.

B. Maximum Variation From Unit to Adjacent Unit: 1/32 inch.

C. Maximum Variation from Plane of Wall: 1/4 inch in 10 ft and 1/2 inch in 20 ft or more.

D. Maximum Variation from Plumb: 1/4 inch per story non-cumulative; 1/2 inch in two stories or more.

E. Maximum Variation from Level Coursing: 1/8 inch in 3 ft and 1/4 inch in 10 ft; 1/2 inch in 30 ft.

F. Maximum Variation of Joint Thickness: 1/8 inch in 3 ft.

3.13 CUTTING AND FITTING

A. Cut and fit for chases, pipes, conduit, sleeves and grounds. Coordinate with other sections of work to provide correct size, shape, and location.

B. Obtain approval prior to cutting or fitting masonry work not indicated or where appearance or strength of masonry work may be impaired.

3.14 CLEANING

A. Clean work under provisions of 01 70 00.

B. Remove excess mortar and mortar smears as work progresses.

C. Replace defective mortar. Match adjacent work.

D. Clean soiled surfaces with cleaning solution.

E. Use non-metallic tools in cleaning operations.

3.15 PROTECTION OF FINISHED WORK

A Protect finished Work.

B. Without damaging completed work, provide protective boards at exposed external corners which may be damaged by construction activities.

3.16 SPECIAL INSPECTION

A. Provide Masonry Special Inspection where required as indicated on project plans.

B. Provide Masonry Special Inspection if indicated on project plans, where required in accordance with the 2018 Edition of the International Building Code.

END OF SECTION